Linear Algebra

Foundations of Mathematical Computing

LearnAI

November 17, 2025

What is Linear Algebra?

Definition

Linear algebra is the branch of mathematics that studies **vectors**, **vector spaces**, and **linear transformations** between these spaces.

Key Concepts

- **Linearity**: Operations that preserve addition and scalar multiplication
- **Vector Spaces**: Collections of objects that can be added and scaled
- **Linear Transformations**: Functions that preserve vector space structure

Why Linear Algebra?

- Foundation for advanced mathematics

Vectors

A vector is an ordered list of numbers:

```
KaTeX error: node:internal/modules/cjs/loader:1424
  throw err;
  ^

Error: Cannot find module 'katex'
Require stack:
- /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js
    at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15)
    at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19)
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22)
    at Module._load (node:internal/modules/cjs/loader:1227:37)
    at TracingChannel.traceSync (node:diagnostics_channel:328:14)
    at wrapModuleLoad (node:internal/modules/cjs/loader:245:24)
    at Module.require (node:internal/modules/cjs/loader:1504:12)
    at require (node:internal/modules/helpers:152:16)
    at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15)
    at Module._compile (node:internal/modules/cjs/loader:1761:14) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    '/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'
  ]
}

Node.js v24.11.1
```

- **Addition**: KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1
- **Scalar Multiplication**: KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1
- **Dot Product**: KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require

Matrices

A matrix is a rectangular array of numbers:

```
KaTeX error: node:internal/modules/cjs/loader:1424
  throw err;
  ^

Error: Cannot find module 'katex'
Require stack:
- /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js
    at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15)
    at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19)
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22)
    at Module._load (node:internal/modules/cjs/loader:1227:37)
    at TracingChannel.traceSync (node:diagnostics_channel:328:14)
    at wrapModuleLoad (node:internal/modules/cjs/loader:245:24)
    at Module.require (node:internal/modules/cjs/loader:1504:12)
    at require (node:internal/modules/helpers:152:16)
    at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15)
    at Module._compile (node:internal/modules/cjs/loader:1761:14) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    '/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'
  ]
}

Node.js v24.11.1
```

- **Addition**: `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1`

- **Multiplication**:

`KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1`

Applications of Linear Algebra

Computer Graphics

- **3D Transformations**: Rotation, scaling, translation
- **Projection**: Converting 3D scenes to 2D displays
- **Animation**: Smooth interpolation between positions

Machine Learning & AI

- **Neural Networks**: Weight matrices and activation functions
- **Principal Component Analysis**: Dimensionality reduction
- **Recommendation Systems**: Matrix factorization techniques

Applications (continued)

Engineering & Physics

- **Circuit Analysis**: Solving systems of linear equations
- **Structural Engineering**: Stress and strain calculations
- **Quantum Mechanics**: State vectors and operators

Economics & Finance

- **Portfolio Optimization**: Risk and return analysis
- **Input-Output Models**: Economic system modeling
- **Linear Programming**: Resource allocation problems

Data Science

- **Image Processing**: Pixel matrices and filters
- **Signal Processing**: Fourier transforms
- **Statistics**: Regression and correlation analysis

Summary

Key Takeaways

✓ Linear algebra provides fundamental tools for mathematical modeling

✓ Vectors and matrices are versatile structures for organizing data

✓ Linear transformations preserve important mathematical properties

✓ Applications span across technology, science, and business

Essential Skills

- Understanding vector operations and geometric interpretation
- Matrix multiplication and properties
- Solving systems of linear equations
- Recognizing linear patterns in real-world problems

Next Steps

- Practice matrix computations
- Explore eigenvalues and eigenvectors
- Study advanced topics like singular value decomposition